"Empowerment through quality technical education"

Ajeenkya DY Patil School of Engineering

Dr. D. Y. Patil Knowledge City, Charholi Bk., Via. Lohegaon, Pune-412 105.

Department of Computer Engineering

Assignments **Form No. IQAC/18a1**

Class: SE Div: C	Course Name: Data Structure	Course Coordinator: Prof.Ishwar Bharambe
Assignment No.4	Assignment based on Units: 4	Date of issue: 14/10/2026

CO 4: Analyze different hashing techniques and collision resolution strategies.

Sr. No.	Questions	Level of Mapping and Number			Blooms Level
		CO	PO	PSO	
Q.1.	Define the following terms with suitable examples: Hash table, hash function, collision, bucket, probe, synonym, overflow.	CO4	PO 1	PSO 1	1
Q.2.	Explain the concept of open hashing and closed hashing with neat diagrams. Discuss their advantages and disadvantages.	CO4	PO1, PO9	PSO 1	2
Q.3.	Differentiate between open addressing and separate chaining as collision resolution techniques. Also, list suitable situations where each technique is preferable.	CO4	PO1-2	PSO 1-2	2
Q.4.	Explain in details linear probing and quadratic probing with example	CO4	PO3,5	PSO 1-2	3
Q.5.	Analyze the working of extendible hashing and discuss how it overcomes the limitations of static hash tables in terms of storage utilization and collision handling.	CO4	PO 2, PO3,5	PSO1,2,3	6

Sign of Course Coordinator: Prof.Ishwar Bharambe	Sign of HoD

